CO-1-AT3

NAME : K. VISHWESH
Reg no : 192511383
Course code : CSA170
C. Name : Artificial Intelligence

Q1) Smart Home Automation system :-

Problem breakdown :-

=> Scenario : A Smart home Automation system manages lighting, temperature, and security based on user behaviour and Environmental condition.

=> Inputs : Real time inputs including motion detection, temperature changes, and time of day. along side learning user Preferences over time.

(i) Explanation of Intelligent Agent Types :-

=> Simple Reflex : select actions based only on the current Percept, ignoring the history of Percepts. They operates using condition - action rules (Eg : if motion is detected, turn on the light).

=> model - Based : Handle partial observability by maintaining an internal state that tracks aspects of the world that can no longer be seen right now. They require a model of how world works

=> Goal : Combine state information with Explicit goal information with Explict goal information to Choose actions that achieve the desired goal.

(Eg. reaching a target temperature).

=> utility - based Agent: use a utility function to desirable a Particular state is allowing the system trade off Competing goals and maximize overall happiness (or) efficiency.

(Eg. Balancing maximum Comfort with minimum Energy Consumption).

(ii) Analysis of Function - within the System & Justification

model Based: useful for tracking whether a room was recently occupied (or) understanding a room's thermal insulation over time.

Goal based: Effective for driving specific outcome like "achieve 22°C by 7:00 PM",

A utility - based framework allows the system to Evaluate trade off

(Eg: Lowering the AC slightly to save Energy while keeping comfort high).

Robot in an unknown maze:-

Problem Breakdown:-

scenario: A robot is placed in an unknown maze and must find a path to the exit without Prior Layout knowledge

ii) Explanation of uniformed search :-

=> UCS: Expands the node with the depth st searche's searche's completeness by repeatedly running depth - limited searches with increasing depth limits.

=> uniform Cost Search :-

=> Time Complexity: $O(b^{c^*/\epsilon})$. where C^* is the optimal cost and ϵ is the lowerbound of step cost.

=> space Complexity: $O(b^{c^*/\epsilon})$ because it keeps all generated nodes in memory.

=> Iterative Deeping Searches :-

=> Time Complexity: $(O(b^d)$ where d is the depth of the shallowest solution.

8. (ii) Agent Design, Decision-making, and Final Recommendation.

=> Agent Relationship: The robot acts as a goal based agent where the goal is finding the exit. Its sensors map the immediate unknown maze Layout into an internal State representing to guide search algorithms.

=> Recommended:-

A goal based agent utilizing Iterative deepening search is effective when memory is Constrained, though if path-cost vary significantly, UCS is Preferred IDS Prevents the robots from getting stuck in infinite paths while keeping memory usage Strictly limited in an unknown Environment.